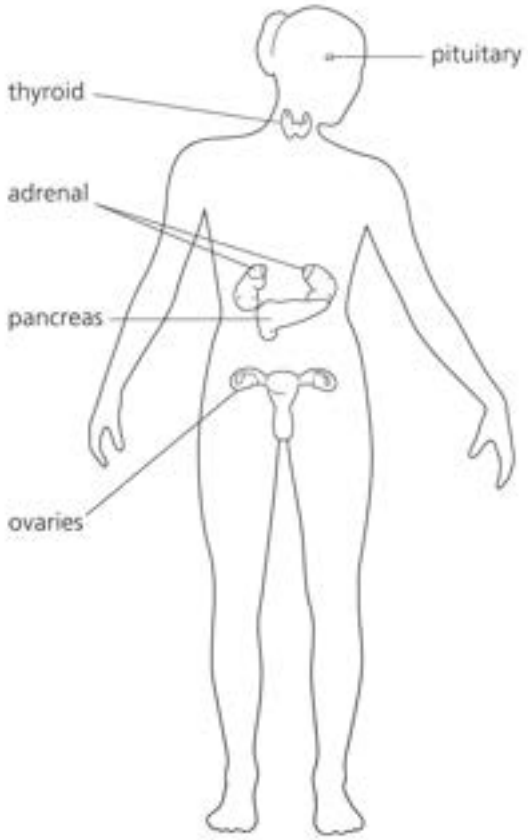
Hormones in the body

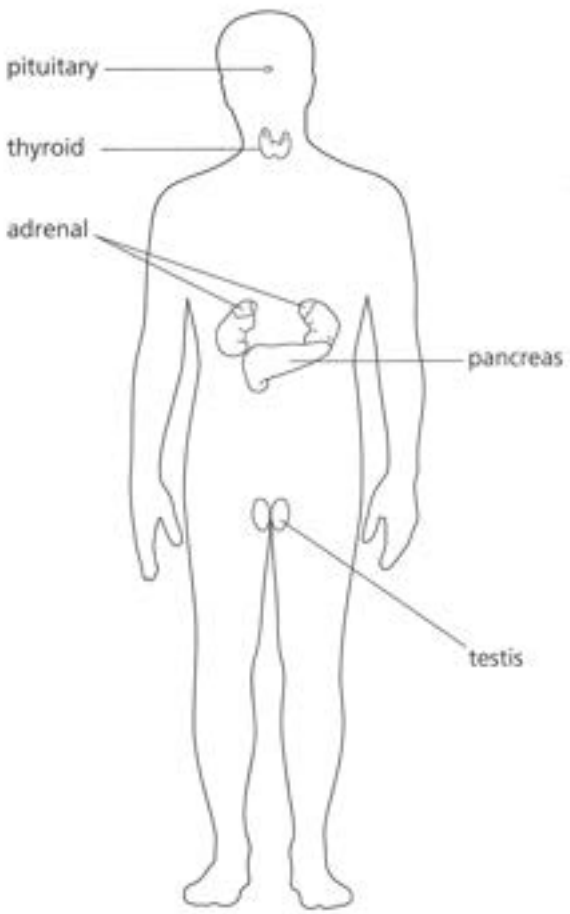
9a

Refer to the labelled diagrams below offering information about hormones and glands in the human male and female body.



Female

Pancreas Special cells found in the pancreas, called Langerhans, produce the hormone called insulin.
Insulin lowers the level of glucose in the blood, causing the liver and muscle cells to convert glucose to glycogen.
Pituitary gland is a very small gland, about the size of a pea. It produces human growth hormone which regulates growth in children.
Thyroid gland produces a hormone called thyroxine, which controls energy release in tissues.
Ovaries produce the sex hormones called oestrogen and progesterone which work the menstrual cycle and also develop the female body.
Adrenal gland produces adrenaline, which prepares the body for action by targeting the muscles, heart, liver and blood vessels.



Male

Male

Pituitary gland
- found at base of brain
- master gland
- controls all other glands
- lack of growth hormone could stunt growth permanently

Pancreas
- produces insulin
- not enough or too much insulin results in diabetes

Thyroid gland
- located in neck
- produces thryoxine
- controls metabolism (the rate at which tissues use energy)
- lack of iodine can result in a disease called goiter (enlarged thryroid)

Testis
- produces sex hormone, testosterone
- controls sexual function throughout the male body
- develops male body characteristics

Adrenal gland
- found near top of each kidney
- adrenalin is produced when stressed or excited
- adrenalin causes increase in heart beat, breathing, blood pressure and sugar levels in the blood

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Hormones in the body

9b

1 Use the information in the diagrams to complete the summary table below.

Name of gland	Location in body	Name of hormone	Name of target organs	Function

2 Refer to your table and to the diagrams on the previous page and write reports on any three glands.

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